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Composition optimization and deformation behavior of FeNiCrCoCu based high entropy alloys at elevated temperatures

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ABSTRACT

This work proposed a materials design strategy combining deep learning (DL), experimental investigations and molecular dynamics (MD) simulations to search for high entropy alloys (HEAs) with high strength at elevated temperature in a model Fe-Ni-Cr-Co-Cu alloy system. The developed DL model evaluated the critical role of element selection in optimizing the performance of FeNiCrCoCu HEAs and contribute to the broader understanding of how multi-element alloys can be tailored for high temperature field. Fe1.1NiCr1.2CoCu0.9 HEAs was designed and fabricated for experimental microstructure characterization and mechanical testing. The microstructure studies showed a dendritic structure containing Fe, Cr, Ni and Co, Cu segregation was detected at the interdendritic regions. Mechanical tests showed the strength of 790 MPa and 446 MPa at 400 and 600 °C. Microcracks initiated at Cu-rich phases and propagated along GBs, resulting in quasi-cleavage fracture. Dynamic recovery occurred at 400 °C and CDRX grains nucleated at 600 °C. Deformation at 400 °C was dominated by dislocation storage, SF accumulation, and strong Cu-rich barrier effects. At 600 °C, the alloy exhibited reduced defect density, weaker phase barrier effects, and more pronounced recovery, shifting the balance toward softening rather than strain hardening. Then MD simulations explored the mechanical behavior at atomic scale, providing detailed insights into dislocation nucleation, grain boundary sliding, and phase transformations. This study contributed to the multi-scale understanding of deformation mechanisms at elevated temperatures.

1. Introduction

High entropy alloys (HEAs) have attracted extensive attention in recent years due to their unconventional design principles and superior mechanical properties, especially under extreme conditions at high temperature [1–5]. Unlike traditional alloys, which are generally composed of one or two primary elements, HEAs incorporate multiple principal elements in near-equiatomic ratios, creating unique, stable microstructures with promising thermal and mechanical stability [6]. The FeCoNiCu-, FeNiCrCoCu-based HEAs system, in particular, has demonstrated exceptional potential for high-temperature applications due to its impressive strength, hardness, and resistance to oxidation [7–10]. Research into FeNiCrCoCu-based HEAs offers the advantage of expanding fundamental knowledge on HEA behavior and developing practical materials for demanding environments, such as those in aerospace, power generation, and automotive industries. The combination of these alloy elements allows for a tailored alloy design with balanced mechanical properties, while the high entropy effect promotes a single-phase, solid-solution structure that enhances high-temperature stability.

In general, the crystal structure, microstructure, and mechanical properties of HEAs are heavily dependent on their compositions. Slight adjustments to the composition ratio can enhance the mechanical properties of the alloy. The critical challenges in FeNiCrCoCu-based HEAs are the design procedure which is time consuming and labor-intensive. To expedite this process, artificial intelligence (AI) has emerged as a powerful tool, offering capabilities to analyze vast datasets, optimize alloy compositions, and predict materials properties with high accuracy [11–15]. For example, Zhao et al. [16] proposed an active learning cycle system to explore HEAs with ultra-high microhardness and unexpected low density. Huang et al. [17] proposed a machine learning (ML) strategy with integrated materials knowledge, including thermodynamics as well as atomic properties, for predicting the ductility of BCC HEAs and providing an insight into plasticity mechanisms.

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Zeng et al. [18] present a comprehensive study on the identification of eutectic high entropy alloys (EHEAs) through integration of CALculation of PHAse Diagrams (CALPHAD), ML, and experimental data, elucidating the dual roles of configurational entropy in phase stabilization and melting-point depression. Besides, our preliminary work [19,20] have demonstrated the superiority of AI in the design of HEAs for enhancing hardness and mechanical properties. Through ML algorithms and data-driven modeling, it enables researchers to quickly identify optimal material characteristics, reducing the time and resources traditionally required for experimental trials. As a result, AI has become an invaluable asset in alloy design, advancing the development of materials with targeted properties for specific applications.

Understanding the mechanical properties and deformation behavior of HEAs is crucial, as it provides insights into their performance under various conditions. Despite traditional theories and experimental methods can provide valuable insights, molecular dynamics (MD) simulations offer a unique perspective by allowing us to observe the behavior of HEAs at the atomic scale [21,22]. Incorporating MD simulations into the study of HEAs adds a powerful dimension to our understanding of their properties. Up to now, a number of studies have employed MD simulations to study the mechanical behavior of HEAs. In particular, some studies have been reported on FeCoCrNiCu-based HEAs [23–27]. Wang et al. [23] investigated the effect of temperature, Cu element content and strain rate on the tensile property of FeCoCrNiCu HEAs. He et al. [24] performed the MD simulations to study the defect pairs, evolution of defect clusters, and recombination rate in a series of alloys. The results showed that FeCoCrNiCu HEAs possessed the highest recombination rate and can be attributed to the long thermal peak life. Cao et al. [26] used MD simulations to research the relationship between mechanical properties and microstructural evolution in FeNiCrCoCux HEAs under uniaxial tension. The results showed that Cu concentration, temperature and strain rate can affect the multiple deformation mechanisms such as dislocation movement and shear strain. To understand the role of Cu, S. Zouaoui et al. [28] investigated the compositional effects of Cu on the structural evolution and mechanical response of nanograined FeNiCrCoCux HEAs. The results showed that low concentrations of Cu could enhance the strength through solid solution strengthening and stabilize the FCC matrix by increasing generalized stacking fault energies (GSFE). While, higher Cu contents (>5 at. %) could promote the grain boundary segregation, grain refinement, and local chemical disorder, leading to GB weakening and mechanical softening. Given the current studies, MD simulations can provide a bridge between theory and experiment, enabling us to probe the intricate atomic interactions that govern the macroscopic behavior of HEAs.

This study systematically performed the composition optimization of FeNiCrCoCu-based HEAs with the goal of improving high-temperature strength. The experimental characterization techniques were employed to elucidate phase distribution, grain structure, and elemental composition. Additionally, mechanical testing was conducted at various elevated temperatures to quantify compressive strength to assess the material suitability for high-temperature applications. The results revealed the intricate interplay between element-strength contributions and overall alloy stability at high temperatures. Then MD simulations were carried out to explore the mechanical behavior under compression of the designed alloy, which can help us to better understand the deformation process.

2. Methodology

This study involves a combined research of deep learning (DL), experimental characterization, and MD simulations for alloy composition design, mechanical property optimization, and deformation mechanism. The schematic flowchart was plotted in Fig. 1(a), and the details of each part were introduced in the following sections.

2.1. Machine learning modeling

In this study, DL techniques were employed to develop predictive models for the properties of HEAs. The methodology involved several stages: data acquisition and preprocessing, model construction and evaluation, inverse design.

Data acquisition and preprocessing: A comprehensive dataset was sourced from published literature on HEA systems [29–32], particularly focusing on alloys containing Co, Cr, Cu, Fe, and Ni. The dataset included compositions along with mechanical property outputs. Then the preprocessing steps involved data cleaning, handling missing values, and normalization of features to ensure consistency and enhance model accuracy.

ML evaluation: The general regression neural network (GRNN), support vector machines (SVM), deep learning (DL) model were trained on a subset of the data using cross-validation techniques to minimize overfitting. Hyperparameter tuning was conducted to optimize model performance and reduce error. The detailed introduction of these three models is as follows.

(1) General Regression Neural Network (GRNN): GRNN is a type of neural network commonly used for regression and function approximation tasks. It is based on nonparametric statistical models and has a one-pass learning algorithm, making it computationally efficient. GRNN uses radial basis functions (RBFs) to estimate the output, which makes it suitable for small datasets with noisy data. GRNN estimates the output (y) based on the probability density function of input–output pairs. The output of GRNN is computed as:

$$y(x) = \frac{\sum_{i=1}^{n} y_i \exp\left(-\frac{\|x-x_i\|^2}{2\sigma^2}\right)}{\sum_{i=1}^{n} \exp\left(-\frac{\|x-x_i\|^2}{2\sigma^2}\right)} \tag{1}$$

where, x is Input vector, x_i is the training data points, y_i is the corresponding output values, σ is the smoothing parameter (spread) and n is the number of training samples.

(2) Support Vector Machines (SVM): SVM is a supervised machine learning model used for classification and regression tasks. It aims to find a hyperplane in a high-dimensional space that best separates data points or fits a function. SVM finds the optimal hyperplane for classification or regression by maximizing the margin or minimizing an error function. In Support Vector Regression (SVR), it seeks a function f(x) within an ϵ-tube to minimize deviations. The objective function for SVR is

$$\min \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n}(\xi_i + \xi_i^*)^2 \tag{2}$$

Subject to:

$$\begin{aligned} &y_i - (w \cdot \phi(x_i) + b) \le \epsilon + \xi_i, \\ &(w \cdot \phi(x_i) + b) - y_i \le \epsilon + \xi_i^*, \\ &\xi_i, \xi_i^* \ge 0 \end{aligned} \tag{3}$$

where, w is the weight vector, $\phi(x_i)$ is the feature mapping function, b is the bias term, C is the regularization parameter controlling the trade-off between margin width and error tolerance, ξ_i, ξ_i^* is the slack variables for handling points outside the ϵ-tube.

(3) Deep Learning (DL) : DL is essentially the process of approximating complex nonlinear mappings using a hierarchy of parametric nonlinear functions. DL models are advanced neural networks capable of automatically learning complex patterns and hierarchical representations from large datasets. They typically have multiple hidden layers, making them suitable for solving problems involving unstructured data. A feedforward neural network (MLP) can be expressed as

$$h^{(l)} = f(W^{(l)}h^{(l-1)} + b^{(l)}), l = 1, 2, \ldots, L. \tag{4}$$

The final output is:

$$\hat{y} = f^{(L)}(W^{(L)}h^{(L-1)} + b^{(L)}) \tag{5}$$

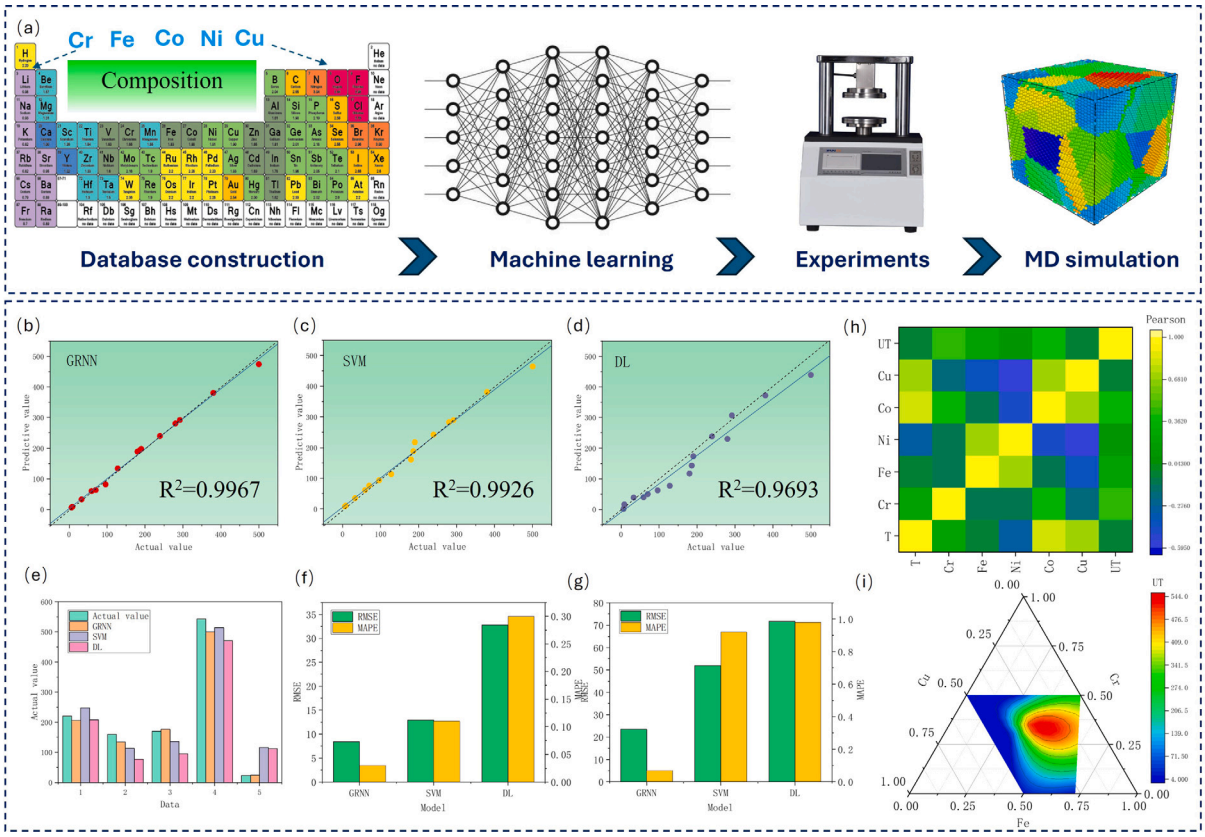


Fig. 1. (a) Schematic flowchart of this work; (b)–(d) ML training results; (e) ML testing results; (f) RMSE and MAPE on training set; (g) RMSE and MAPE on testing set; (h) Effect of temperature and composition on strength; (i) Effect of (Fe, Cr, Cu) by inverse design.

The core of training deep learning models is gradient descent and backpropagation. The goal of network training is to minimize the empirical loss:

$$\min_{\theta} L(\theta) = \frac{1}{N}\sum_{i=1}^{N} \ell(f(x_i;\theta), y_i) \tag{6}$$

For example, the mean squared error (MSE) for regression tasks is

$$L = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{7}$$

The backpropagation algorithm applies the chain rule to efficiently compute the gradient of the loss function with respect to network parameters. Gradients of the loss with respect to parameters (W, b) are calculated:

$$\frac{\partial L}{\partial W} = \frac{\partial L}{\partial a}\cdot\frac{\partial a}{\partial z}\cdot\frac{\partial z}{\partial W} \tag{8}$$

The optimization algorithm is using stochastic gradient descent (SGD):

$$W \leftarrow W - \eta\frac{\partial L}{\partial W} \tag{9}$$

where, η is the learning rate.

Model evaluation:Model accuracy was assessed for robustness and generalizability using metrics such as mean average percentage error (MAPE), root mean squared error (RMSE), and R-squared (R^2) scores [33].

$$MAPE(\%) = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \times 100\% \tag{10}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{11}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{12}$$

where, y_i is the experimental value, $\hat{y}_i$ is the predicted value, $\bar{y}$ is the average value of y_i.

Prediction and inverse design: The constructed DL model can provide accurate predictions. By principal component analysis (PCA), it can help us understand how individual features influenced the predicted mechanical properties and identified the most impactful features. Then the appropriate composition ratio can be obtained in the expect of obtaining desirable mechanical properties.

2.2. Experimental method

Fe1.1NiCr1.2CoCu0.9 HEAs were synthesized using vacuum arc-melting. The raw elemental metals, each with a purity exceeding 99.9%, were melted in a water-cooled copper mold under a high-purity argon atmosphere to prevent oxidation. To enhance chemical homogeneity, the melting process was repeated a minimum of five times, ensuring uniform distribution of elements throughout the alloy. This rigorous process yielded high-quality HEA samples, suitable for subsequent analyses of their microstructure and mechanical properties.

The phase types of the prepared HEAs were detected using X-ray diffraction (XRD) with Cu target. The samples for the microstructure observation were carefully mechanical ground and polished. The microstructures of the HEAs were observed using scanning electron microscopy (SEM) and Electron backscatter diffraction (EBSD), using EDAX software and a TSL OIM EBSD system. Compression tests were carried out at 400 and 600°C using AG-X Plus 250 kN/50 kN Electronic Universal Testing Machine with a strain rate of 2 mm/min. The deformed microstructure was examined by using transmission electron

microscope (TEM). Thin foils were prepared from the middle of the deformed sample. The selected-area-electron-diffraction (SAED) analysis and high-resolution TEM (HRTEM) observations were performed on the sample.

2.3. Simulation details

The polycrystalline FCC- Fe1.1NiCr1.2CoCu0.9 HEAs model was established to study the uniaxial plastic deformation, with a lattice spacing of 3.62 Å and a size of 100 Å × 100 Å × 100 Å (length × width × height, respectively). The crystallographic orientations of the model are: x: [100], y: [010], z: [001]. In this study, the mechanical behavior of the model was examined to explore the plastic deformation mechanism. The embedded atom model (EAM) potential [34] was employed for FCC Fe1.1NiCr1.2CoCu0.9 HEAs. In all simulations, the NPT ensemble [35] was applied with a fixed time step of 1 fs. Periodic boundary conditions were set along the X- and Y-directions, while the Z-direction was left with a free boundary condition. Prior to deformation, the system was energy minimized by controlling temperature and pressure with the Nose–Hoover thermostat and barostat [36], respectively. Motion equations were integrated using the velocity-Verlet algorithm [37]. The model was then relaxed for 50 ps at a designated temperature to establish thermodynamic equilibrium. During the compressive deformation, strain rate of 5×10^{-9} s^{-1} was applied along the X-direction, and different temperature conditions were evaluated (673 K and 873 K).

All simulations were performed using the Large-scale Atomic/Molecular Massively Parallel Simulator (LAMMPS) [38]. The structural analysis and visualization were performed using OVITO [39], with the dislocation extraction algorithm [40] and common neighbor analysis (CNA) [41].

3. Results and discussion

3.1. Machine learning results

Fig. 1(b)–(d) presented the regression analysis on the training results by GRNN, SVM and DL, respectively. From the predicted results by GRNN and SVM, as shown in Fig. 1(b) and (c), it can be observed that the data points were evenly distributed near the asymptotes. The asymptotes agreed well with the fitted lines. While, from the regression analysis on the training set by DL model (Fig. 1(d)), there is a significant deviation between the data points and the asymptotic line. There is also a significant deviation between the asymptotes and the fitted lines. Through statistical analysis on actual value and predicted value, the R^2 value was calculated as 0.9967, 0.9926 and 0.9693 for GRNN, SVM and DL model, respectively. To verify the reliability of the trained model, Fig. 1(e) presented the testing results on unseen data. It is intuitive to see that the actual value matches the predicted value very well, while there is a significant gap between the predicted values by DL model and the actual values. The R^2 value regarding testing set was 0.9955, 0.9146 and 0.8655 for GRNN, SVM and DL model, respectively, indicating that the GRNN and SVM model performed better than DL model. To further evaluate the performance of each model, Fig. 1(f) and (g) summarized the RMSE and MAPE on the training and testing sets, respectively. In general, the smaller value of RMSE and MAPE indicated the higher prediction accuracy. It can be deducted that GRNN model exhibited lower RMSE and MAPE than SVM and DL, not only on training set but also on the testing set. GRNN model has been demonstrated to possess high superiority in modeling small sample data. Given that, the superior predictive performance of GRNN model demonstrated its ability to provide further prediction.

Fig. 1(h) plotted the effect of elements and temperature on the strength by calculating the Pearson correlation coefficient. It ranges between −1 and 1, which can reflect the degree of correlation between input and output variables. The strength decreases with the increase of temperature, thus showing a significant negative correlation between temperature and strength. Among the elements, there is a positive correlation between (Fe, Cr, Ni, Co) and strength, while there is a negative correlation between Cu and strength. According to the sensitivity analysis, the degree of elemental influence can be derived: Cr>Co>Fe>Ni. Regarding the role of Cu, the obtained trend agreed well with the findings in [28], where high Cu content can lead to excessive elemental segregation. Considering the high cost of Co and minor effect of Ni, the content of these two elements are not change when designing alloys. Hence, increasing the content of Cr and Fe, and decreasing the content of Cu could improve the strength of FeNiCrCoCu-based HEAs. Recognizing this, Fig. 1(i) presented the ternary phase diagram to illustrate the effect of (Fe, Cr, Cu) on the strength. Given this analysis, Fe1.1NiCr1.2CoCu0.9 HEAs were designed for further investigation.

Table 1
Composition of different regions in the as-cast Fe1.1NiCr1.2CoCu0.9 HEAs.

Location	Chemical composition (at.%)				
	Cr	Fe	Co	Ni	Cu
A	24.83	22.82	21.37	21.42	9.56
B	22.33	21.06	18.90	21.75	15.97
C	3.25	4.05	3.99	7.81	80.89

3.2. Microstructural characterization

Fig. 2(a) shows the XRD pattern obtained from the Fe1.1NiCr1.2CoCu0.9 HEAs. It can be seen that there are two peaks corresponding to two FCC phases. In each pair of peaks, the peak with lower intensity represents the Cu-rich phase, which has a larger lattice constant than the matrix phase. Additionally, the higher intensity was detected to the matrix FCC phase, which may be attributed to its larger grain sizes compared to the Cu-rich phase. This is similar to the CoCrCuFeNi HEA reported in [30]. Fig. 2(b) and (c) shows the microstructure of the as-cast Fe1.1NiCr1.2CoCu0.9 HEAs, identifying typical dendrite phases. The average width of dendrites was ~ 20 μm. Energy dispersive X-ray spectroscopy (EDS) was measured and the chemical composition of different regions (A, B and C) in the alloy was listed in Table 1. It is obvious that region A was poor in Cu element, and the matrix phase was composed of nearly nominal contents of Co, Cr, Fe and Ni. The content of Cu element increased in the area near the interdendritic region (region B). The significant increase in Cu content in the interdendritic region (region C) indicated the formation of Cu-rich phase. Herein, Cu segregation is anticipated in this alloy due to the high positive enthalpy of mixing among Fe–Cu, Co–Cu, and Cr–Cu [42]. The matrix phase contains only ~10 at.% of Cu, while most of Cu segregates in the interdendritic phase and the content can reach up to ~80 at.%.

Fig. 2(d)–(i) presented the EBSD analysis of Fe1.1NiCr1.2CoCu0.9 HEAs. The grain orientation was revealed by the inverse pole figure (IPF) map in Fig. 2(d), and the grain structure as well as the distribution of grain boundary were presented in image quality (IQ, Fig. 2(e)) map and grain boundary (GB, Fig. 2(f)) map, respectively. In particular, low-angle grain boundaries (LAGBs, 0–15°) were observed in individual grains due to the slight misorientations within a single grain during solidification. From kernel average misorientation (KAM) map, these grains exhibit high KAM values, as shown in Fig. 2(g). The distribution of grain size was summarized in Fig. 2(h), the average grain size of Fe1.1NiCr1.2CoCu0.9 HEAs was 20.3 μm. Fig. 2(i) shows the (100), (110), and (111) pole figures (PFs), indicating the texture intensity of 5.15.

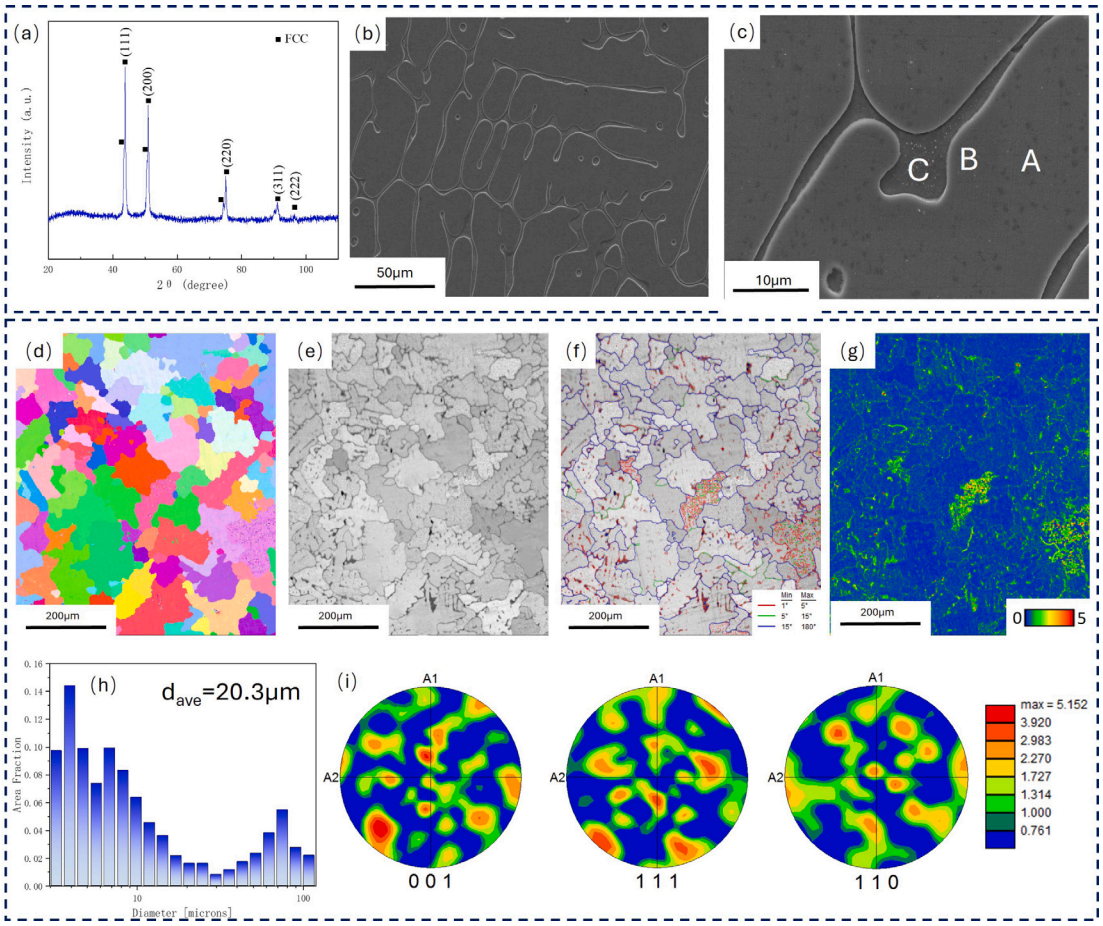


Fig. 2. (a) XRD pattern (b) (c) SEM images of Fe1.1NiCr1.2CoCu0.9 HEAs; EBSD analysis (d) IPF map (e) IQ map (f) GB map (g) KAM map (h) Grain size (i) PF map.

3.3. Mechanical properties

Fig. 3(a) plots the engineering stress–strain curves of Fe1.1NiCr1.2CoCu0.9 HEAs at elevated temperatures of 400 and 600°C. The strength decreased by increasing the deformation temperature. The fracture strength can achieve 790 MPa at 400°C, with a fairly high plastic strain of 48.6%. As the temperature increased to 600°C, the strength decreased to 446 MPa. The work hardening was detected after the flow stress reached the peak. Fig. 3(b) illustrates the corresponding true stress–strain curves, and the strain-hardening rates ($\theta = d\sigma/d\varepsilon$) at different temperatures were plotted in Fig. 3(c). At the initial stage of deformation after yielding, the strain-hardening rate is typically high due to the high dislocation density and interactions [43]. After a rapid drop, it reaches a steady-state flow regime where work hardening and work softening processes balance. As the temperature increased from 400°C to 600°C, strain hardening tends to be reduced due to the enhanced ability of dislocations to overcome barriers and obstacles more easily, often resulting in a lower resistance to plastic deformation [44].

To investigate the deformation mechanisms at elevated temperatures, EBSD analysis was carried out on the deformed sample. Fig. 3(a1)–(d1) plotted the IPF map, IQ map, GB map and KAM maps of the deformed sample at 400°C, respectively. The color gradients can be identified within grains, as shown in Fig. 3(a1), which represents the variations in crystallographic orientation across the grain. When grains undergo plastic deformation, dislocations accumulate and rearrange to create sub-grain structures and small changes in orientation, resulting in a gradual shift in orientation across the grain. Due to the slight misorientations, Fig. 3(c1) indicated that the deformed HEAs contained a huge quantity of LAGBs (58%), which can also be reflected in the misorientation angle distribution map illustrated in Fig. 3(e1). With respect to the deformation at 600°C, as illustrated in Fig. 3(e2), the misorientation angle was mainly concentrated in the LAGBs, while extremely low at the HAGBs. During the deformation process, grains attempted to accommodate strain by realigning their orientations to minimize internal stresses [45], leading to the formation of parallel deformation bands across grains, as shown in the higher magnified IPF map (Fig. 3(a1-2)). This phenomenon has also been observed during the deformation process at 600°C (Fig. 3(a2) and (a2-2)). It was closely related to the Cu-rich phase. The reduced SFE suppressed dislocation cross-slip while promoting stacking fault formation and deformation twinning [46–48]. In KAM map, the color scale represented the local crystal orientation mismatch, where low KAM values indicated the low local strain/dislocation density and high KAM values indicated the severe lattice distortion and increased dislocation accumulation. As shown in Fig. 3(d1)–(d2), the increased local misorientation can be identified along high-angle grain boundaries (HAGBs, > 15°) and near the deformation bands, which are typically associated with high dislocation densities. This indicated significant plastic deformation, where dislocations accumulated due to higher plastic strain.

As the temperature increased to 600°C, some equiaxed grains surrounded by HAGBs were detected as shown in Fig. 3(a2), which demonstrated the occurrence of dynamic recrystallization (DRX). It led the transformation from a highly deformed microstructure to a refined and stable arrangement of new grains that accommodated continued deformation more effectively. In addition, Fig. 3(c2) showed a noticeable increase in amount of LAGBs, to about 91% (Fig. 3(e2)), during compression at 600°C. This was due to the enhanced dislocation mobility and dynamic recovery processes at higher temperatures [49]. The

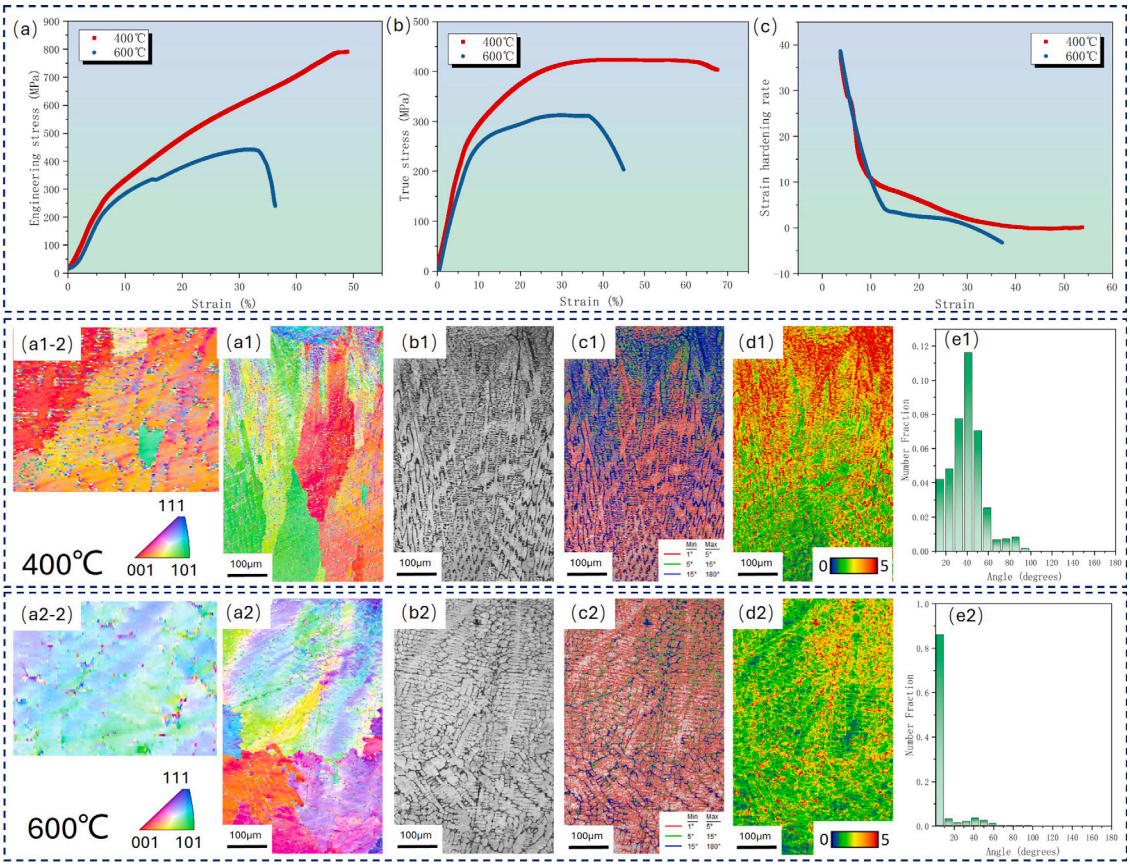


Fig. 3. Mechanical properties of Fe1.1NiCr1.2CoCu0.9 HEAs at elevated temperatures (a) Engineering stress–strain curves; (b) True stress–strain curves; (c) Strain-hardening rate; EBSD analysis on the deformed sample (a1)–(e1) 400°C; (a2)–(e2) 600°C. Herein, (a1)(a2) IPF map, (a1-2)(a2-2) Enlarged view of IPF map, (b1)(b2) IQ map, (c1)(c2) GB map and (d1)(d2) KAM map (e1)(e2) Misorientation angle. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

internal stresses can be relieved by creating stable sub-grains, leading to an increase in LAGBs. This was a typical phenomenon in the early stages of microstructural evolution before complete recrystallization occurred. In addition, the occurrence of DRX at higher temperatures can lead to the reduction in flow stress.

To understand the fracture mode and deformed microstructure, Fig. 4 presented the SEM images of deformed sample accompanied with EDS analysis. The surface morphology (Fig. 4(a1)–(d1)) of Fe1.1NiCr1.2CoCu0.9 HEAs compressed at 400°C revealed the significant plastic deformation accompanied by localized cracking and interface decohesion. In Fig. 4(a1), intergranular cracks were observed along certain GBs, suggesting strain localization and limited GB cohesion under elevated stress. From the EDS analysis (Fig. 4(c1)–(i1)), Cu exhibited clear segregation along GBs and fracture paths, forming Cu-rich segregated region. The segregation of Cu atoms in multicomponent Fe–Ni–Cr–CoCu HEAs was mainly associated with the thermodynamic incompatibility between Cu and other elements. Specifically, Cu exhibited a positive mixing enthalpy with Fe, Cr, and Co, indicating relatively weak atomic interactions among these elements. As a result, the solid solution became thermodynamically less stable when the Cu concentration increases, causing Cu atoms to preferentially aggregate and form Cu-rich regions or secondary phases in order to minimize the overall Gibbs free energy of the system. In addition, Cu possessed a relatively larger atomic radius and lower bonding affinity compared with the other constituent elements, which further promoted the local lattice distortion and chemical inhomogeneity. During solidification, the limited mutual solubility and sluggish diffusion behavior characteristic of HEAs can facilitate elemental partitioning, leading to the formation of Cu-enriched regions. Combined the observation in SEM images, it was deduced that the cracks initiated and propagated along Cu-rich secondary phases. This also indicated the weaker bonding of Cu-rich phase with the Fe–Ni–Cr–Co matrix. In Fig. 4(b1), the Cu-rich phase has undergone severe deformation, indicating intense dislocation slip activity. The morphology of the cracks and fractured regions (Fig. 4(c1) and (d1)) exhibited features of quasi-cleavage fracture, where brittle facets coexisted with localized plastic deformation zones.

After compression at 600°C, the microstructure (Fig. 4(a2)–(i2)) showed a distinctly different deformation morphology compared to 400°C. As seen in Fig. 4(a2), combined with EDS mapping analysis in Fig. 4(c2)–(i2), the overall surface showed a relatively complete structure with fewer long cracks. It indicated the enhanced ductility and improved GBs accommodation at higher temperature. The deformation pattern became more homogeneous, suggesting that thermally activated mechanisms, such as dislocation climb, cross-slip, and recovery, facilitated smoother plastic flow. In Fig. 4(b2) and (c2), GBs remained largely intact, and cracks that appeared were short, blunt, and discontinuous. The Cu-rich phases observed at this temperature tend to be larger due to thermal coarsening, reducing their effectiveness as crack initiation sites. The fracture morphology (Fig. 4(d2)) also showed facets resembling cleavage planes, indicating the quasi-cleavage fracture. These are interspersed with dimples or tearing ridges, indicating some localized plastic deformation before final separation. The crack propagation involves dislocation motion and micro-crack linkage along specific crystallographic planes. Overall, deformation at 600°C proceeded predominantly through dislocation glide and recovery, accompanied by limited interfacial damage and microvoid formation. Overall, deformation at 600°C proceeded predominantly through dislocation glide and recovery, accompanied by limited interfacial damage and microvoid formation.

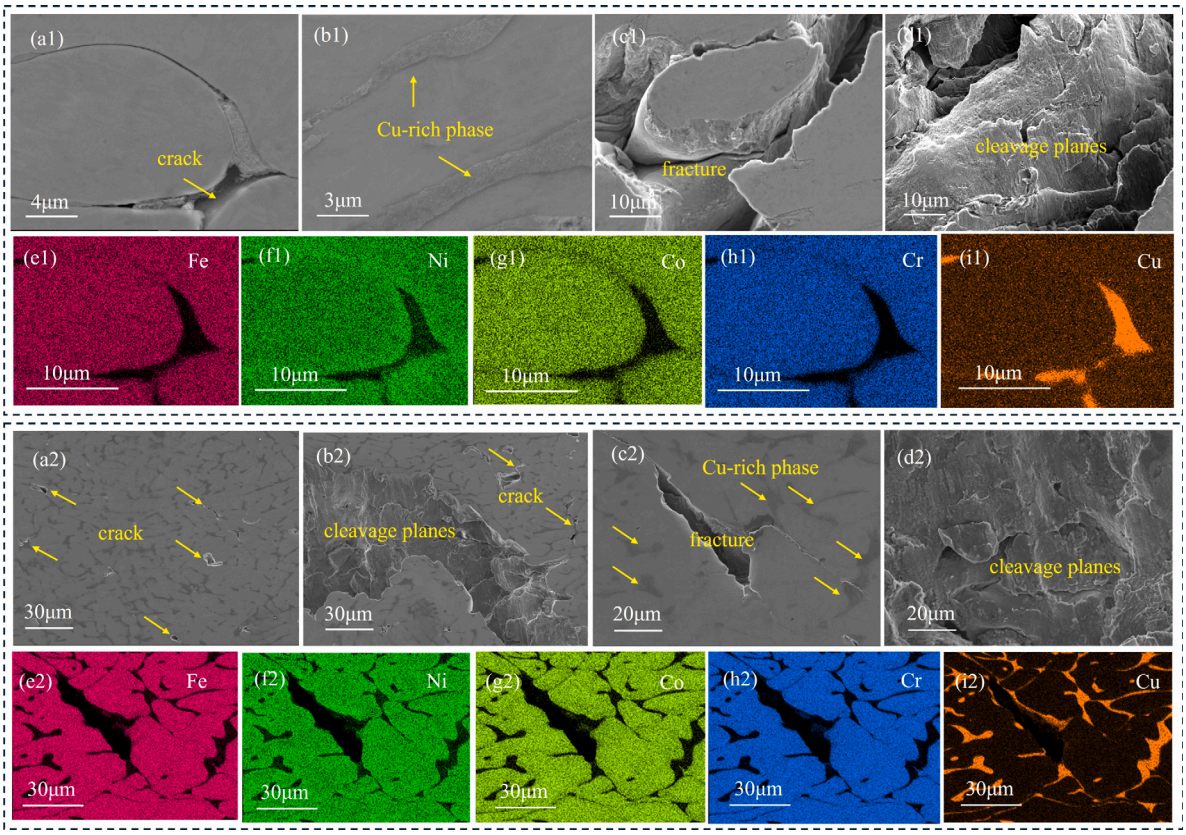


Fig. 4. (a1)–(d1) Deformed microstructure after compression at 400°C; (c1)–(i1) EDS mapping analysis of (a1); (a2)–(d2) Deformed microstructure after compression at 600°C; (c2)–(i2) EDS mapping analysis of (c2).

Consequently, at moderate temperature, the deformation in Fe1.1NiCr1.2CoCu0.9 HEAs is characterized by Cu segregation-induced interface weakening and quasi-cleavage fracture. Microcracks initiated at Cu-rich phases and propagated along GBs, indicating a competition between ductile and brittle mechanisms. As the temperature increased from 400 to 600°C, the enhanced atomic mobility promoted the partial redistribution of Cu atoms, reducing the local compositional gradients and alleviating internal stress concentrations around Cu-rich regions. At the same time, the increased diffusion rate also facilitated the coarsening of Cu-rich phases. Compared with fine and highly localized Cu-rich precipitates, these coarsened regions exhibited lower stress concentration sensitivity and therefore became less effective crack initiation sites. The Fe1.1NiCr1.2CoCu0.9 HEAs exhibited enhanced ductility, reduced interfacial damage, and more uniform deformation. The shift from cracking at 400°C to stable, recoverable plasticity at 600°C highlighted the crucial role of temperature-assisted diffusion and dynamic recovery in improving the mechanical resilience under compressive loading.

To further characterize the deformed microstructure, Figs. 5 and 6 presented TEM characterization on the Fe1.1NiCr1.2CoCu0.9 HEAs deformed at 400 and 600°C, respectively. As illustrated in Fig. 5(a), two representative grains (Grain 1 and Grain 2) were identified. The High-Angle Annular Dark-Field (HAADF) images with elemental EDS maps illustrated in Fig. 5(h)–(m) confirmed the FCC phase in the matrix with Cu-rich secondary phase clearly observed at the GB region between the two grains. This indicated that the Cu-rich phase preferentially segregated to GB regions and interfaces. Many studies have confirmed that the precipitated phase plays an important role in influencing the local mechanical responses and determining the mechanical properties of alloys [50,51]. Moreover, the dislocation cells were detected along GBs. The selected area electron diffraction (SAED) images confirmed the FCC matrix structure (Fig. 5(c)), with additional diffraction spots indicating the presence of a Cu-rich secondary phase (Fig. 5(d)). In Fig. 5(b), the deformed microstructure exhibited a high density of dislocations, forming dislocation walls and substructures. Particularly in Fig. 5(c), a large number of dark particles were identified. The HAADF image, combined with corresponding EDS maps in Fig. 5(n)–(s) showed that Cu was enrich in the secondary phase, while Fe, Ni, Cr, and Co remained more uniformly distributed in the FCC matrix. It confirmed that the dark particles were Cu-rich phases, which distributed in FCC matrix. The high-resolution TEM (HRTEM) images in Fig. 5(f) and (g) showed detailed evidence of defect-phase interactions. The interfaces between FCC matrix and second-phase particles could be observed. The lattice fringes of the Cu-rich precipitate were visible, and lattice mismatch at the matrix/precipitate interface was evident. The stacking faults (SFs) and partial dislocations were clearly resolved near the interfaces and within the FCC lattice, highlighting the role of SF-mediated slip in plasticity. It can be deduced that the interface could act as a strong barrier to dislocation motion, causing pile-ups and promoting local stress concentration [52,53]. Overall, the microstructure demonstrated that the plastic deformation of Fe1.1NiCr1.2CoCu0.9 HEAs at 400°C was mainly governed by the dislocation slip and wall formation in the FCC matrix, providing primary strain accommodation. In the present alloy system, the enhanced strain hardening originated from the combined effects of the FCC matrix/Cu-rich phase interfaces and the local variation in SFE. Cu-rich phase precipitation at GBs, accompanied with the slight formation of SFs, acted as strong barriers to dislocation motion, further enhanced the strain hardening.

Fig. 6(a) showed the overall microstructure deformed at 600°C. Similar to it at 400°C, we can also observe two typical grains. The dislocation density was particularly high, distributing both within grains and along GBs. It can be deduced that dislocations interact more dynamically with precipitates at GBs, with evidence of partial absorption or bypassing rather than persistent pile-ups typical at 400°C. Fig.

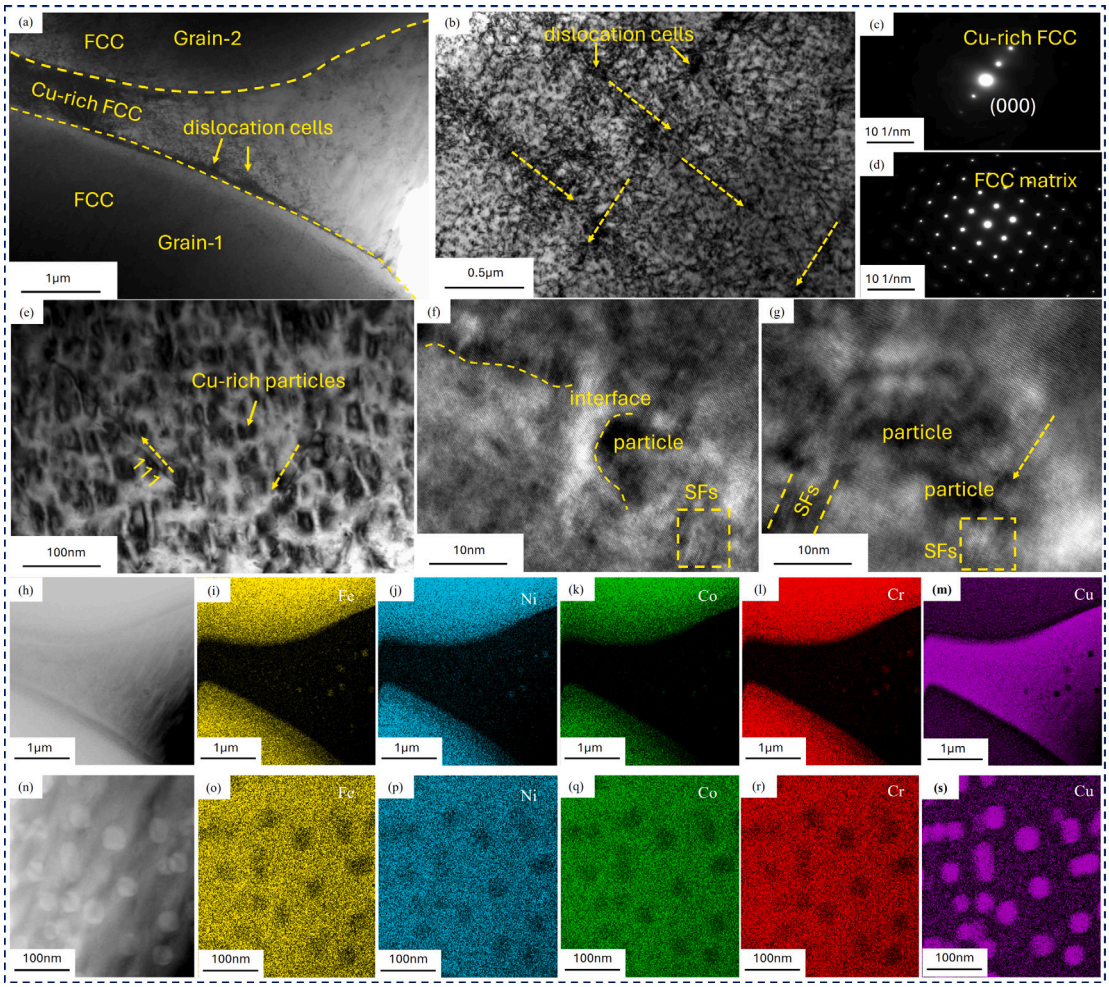


Fig. 5. Deformed microstructure at 400°C: (a)(b)(e) TEM images of deformed microstructure; (c) (d) SAED images; (f) (g) (j) the corresponding HRTEM images; (h)–(m) and (n)–(s) HAADF image with elemental EDS maps.

6(b) showed lower overall dislocation density than at 400°C since enhanced dislocation climb and annihilation promoted partial recovery. The diffraction patterns (Fig. 6(d)) confirmed the persistence of the FCC matrix structure at 600°C. The Cu-rich secondary phase remains detectable in SAED image (Fig. 6(c)), but its diffraction spots appear broader, suggesting partial coarsening and disorder at elevated temperature. From Fig. 5(e), the dislocations formed looser networks and subgrain boundaries rather than dense tangles or walls, more homogeneous dislocation distributions appeared within grains compared with 400°C. At higher resolution (Fig. 6(f)), the FCC lattice also showed extended SFs and deformation twins at interface. Cu-rich precipitates are still embedded at GBs, but the matrix-precipitate interface appears more diffuse, indicating the interfacial relaxation at higher temperature (Fig. 6(g)). The relaxation at higher temperature reduced the lattice mismatch and makes the interfaces less effective as strong dislocation barriers. Moreover, The HAADF image with corresponding EDS mappings reveals continued Cu enrichment at GB regions (Fig. 6(h)–(m)) and secondary phase particles (Fig. 6(n)–(s)). The Cu-rich particles were more chemically segregated, consistent with thermal coarsening. While, Fe, Ni, Cr, and Co remained evenly distributed in the FCC matrix. Compared with 400°C, where fine Cu-rich precipitates acted as strong local obstacles, the coarsened Cu-rich phases at 600°C provided weaker pinning, allowing dislocations to bypass or shear them more readily. In general, the reduced dislocation density was detected during the deformation at 600°C due to thermal recovery, with more uniform distributions and partial annihilation. SFs became less stable and partially annihilated. Cu-rich phases became coarsen and interfaces were relaxed, reducing barrier strength and allowing easier dislocation motion. Hence, the more dynamic recovery and dislocation annihilation at 600°C lowered strain hardening, but improved ductility and homogenize deformation.

To explain the deformation behavior at the atomic scale, MD simulations were conducted at 400 and 600°C, and the results were plotted in Figs. 7 and 8. The physical models used for MD simulation and observation were illustrated in Fig. 7(a). Then, Fig. 7(a1)–(d3) presented the microstructure evolution of Fe1.1NiCr1.2CoCu0.9 HEAs during deformation at 400°C on the (100) crystal plane, revealing the dislocation behavior (Fig. 7(a1)–(d1)), phase transition (Fig. 7(a2)–(d2)), and atomic strain localization (Fig. 7(a3)–(d3)) with the increasing strain. Different types of dislocations have been discovered during the deformation process of alloy, including $1/6\langle 112\rangle$ Shockley type, $1/2\langle 110\rangle$ Perfect type, $1/6\langle 110\rangle$ Stair-rod type, $1/3\langle 100\rangle$ Hirth type and other type. Among the mentioned dislocations, the Shockley type dislocation dominated in deformation, which is a common phenomenon in FCC alloys [54,55]. It can be observed that the dislocations nucleated from GBs and local stress concentration area at lower strain ($\epsilon = 0.12$). As strain increased, dislocations multiplied and glided readily, indicating the thermal activation of slip. As the strain increased, the dislocations developed, interacted, and rearrange into networks and partial walls until the strain of 0.16. At higher strains ($\epsilon > 0.16$), dense dislocation tangles dominated the microstructure. The deformation at 400°C promoted dynamic recovery, resulting in partial annihilation of mobile dislocations, but the overall dislocation density remained high, sustaining work-hardening.

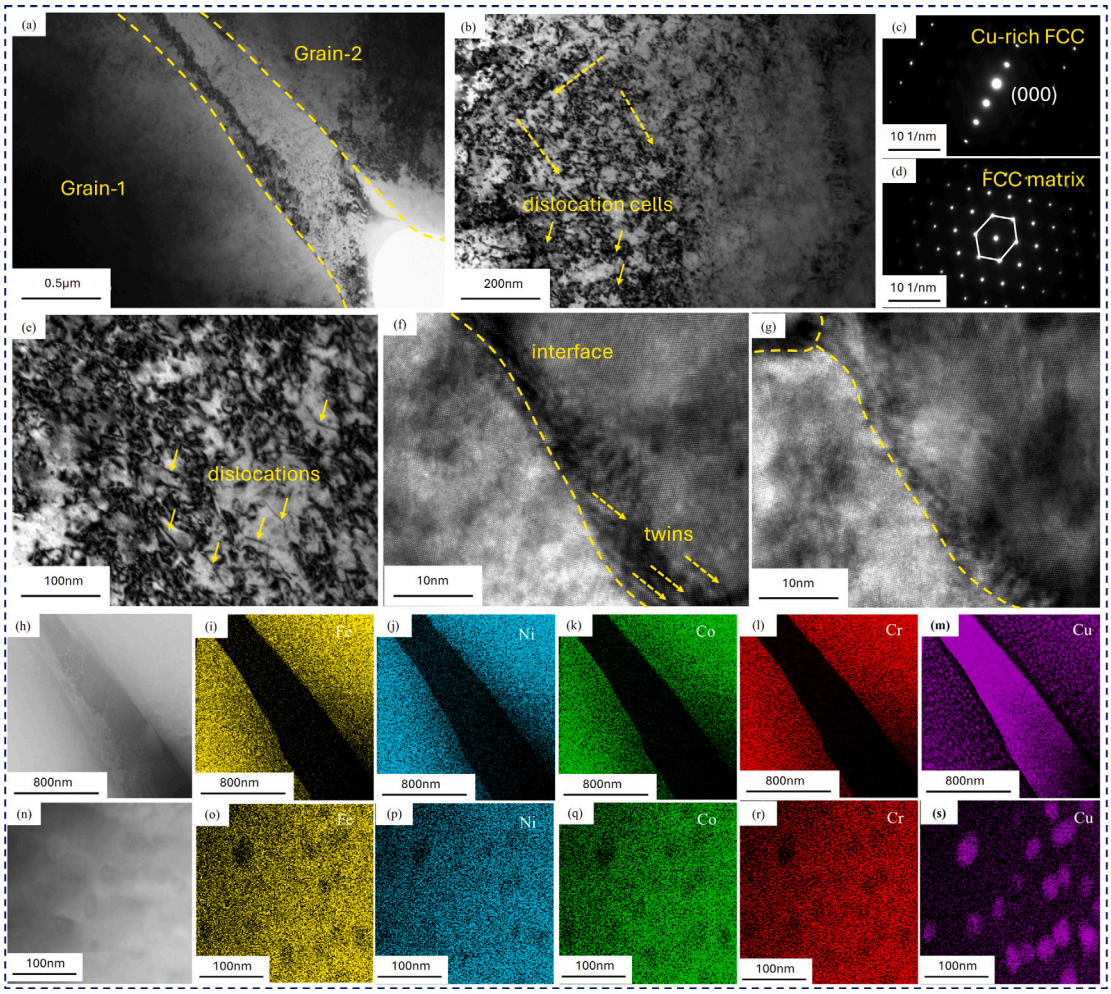


Fig. 6. Deformed microstructure at 600°C: (a) (b) (e) (f) (h) TEM images of deformed microstructure; (c) (d) HAADF images; (g) (i) (j) the corresponding HRTEM images; (k)–(p) EDS maps.

The atomic-scale snapshots in Fig. 7(a2)–(d2) showed a progressive FCC-HCP phase transformation, indicating the formation of SFs. The formation of ISF, ESF and amorphization during compression was summarized in Fig. 7(i). At small strains, a small amount of SFs appear near dislocation cores and only minor fraction developed across the grain. With increasing strain, SFs further expanded, assisted by thermally activated atomic shuffling. At larger strains, SFs lamellae formed and coexisted with dislocation networks, confirming a significant strain-accommodation mechanism under 400°C deformation. Hence, the large-strain deformation generated a higher dislocation density in HEAs, and simultaneously led to complex interactions between the defects of the crystal lattice such as dislocations and SFs.

The corresponding atomic strain distribution were plotted in Fig. 7(a3)–(d3). The strain was localized near dislocation sources and GBs at early stages. As deformation progresses, atomic strain spreads over larger regions, aligning with slip bands. At higher strain, the strain distribution becomes highly heterogeneous, with strong localization along dislocation walls and phase boundaries, indicating the coupling between dislocation motion, SFs thickening, and local lattice transformation.

Moreover, we also captured the microstructure evolution on the (111) crystal plane to understand how dislocations, stacking faults (SFs), and grain boundaries (GBs) collectively evolved with increasing strain (Fig. 8). At $\epsilon = 0.12$ (Fig. 8(a)), dislocations nucleated near GBs and propagate along {111} slip planes, with a few SFs appearing in grain interiors. As the strain increased to 0.14, dislocation density rose, and extended SFs began to develop across grains, indicating the activation of partial dislocations. By $\epsilon = 0.16$ and 0.18, SFs became thicker and more extended.

For better understanding, Fig. 8(a1)–(d1) and (a2)–(d2) plotted the magnified view 1 and view 2 to highlight the evolution of dislocation-GB interactions. GBs acted as dislocation sources, emitting partial dislocations into the grain interior. With the increasing strain, mobile dislocations piled up against GBs and junctions emerged at GBs due to dislocation-GB reactions, leading to the formation of pinned dislocation structures. The GBs were saturated with dislocation clusters, creating severe lattice misorientation and promoting nucleation of additional SFs adjacent to GBs.

To investigate the deformation behavior at 600°C, Fig. 7(e1)–(h3) presented the microstructure evolution of Fe1.1NiCr1.2CoCu0.9 HEAs on the (100) crystal plane. From the dislocation evolution illustrated in Fig. 7(e1)–(h1), the dislocations multiplied rapidly and glided over longer distances compared to the deformation at 400°C due to the enhanced thermal activation of slip. Regarding the phase transition evolution (Fig. 7(e2)–(h2)), the FCC-HCP transformation was observed at all strain levels but is more dynamic at 600°C. An increased density of SFs formed even at small strain, as well as the extended slip bands. The dynamic recovery became significant at larger strain ($\epsilon = 0.18$). The recovery and climb processes dominated the deformation, resulting in a higher density of dislocation structure compared with lower-temperature deformation. Compared to deformation at 400°C, their stability was reduced due to thermal fluctuations and more SFs coexisted with dislocation structures. With the deformation, some partially

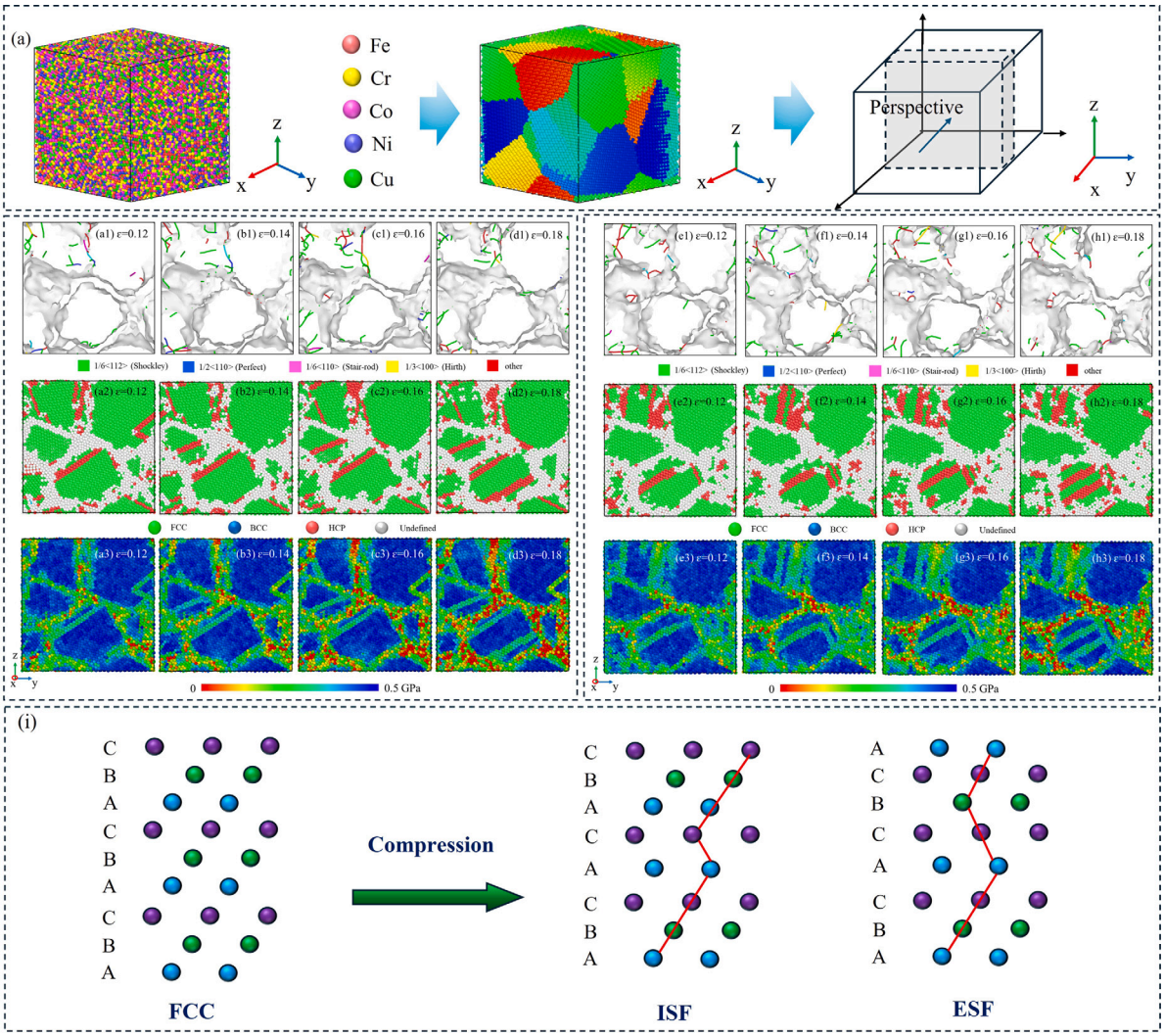


Fig. 7. (a) MD model for simulation and observation; (a1)–(d3) Microstructure evolution of Fe1.1NiCr1.2CoCu0.9 HEAs during deformation at 400°C on the (100) crystal plane: (a1)–(d1) Dislocation evolution, (a2)–(d2) phase transition and (a3)–(d3) atomic strain distribution at different strain; (e1)–(h3) Microstructure evolution of Fe1.1NiCr1.2CoCu0.9 HEAs during deformation at 600°C on the (100) crystal plane: (e1)–(h1) Dislocation evolution, (e2)–(h2) phase transition and (e3)–(h3) atomic strain distribution at different strain;(i) Formation of ISF, ESF and amorphization during compression.

dissolve back into FCC, indicating a competition between deformation-induced transformation and thermally assisted reversion.

It was similar to 400°C, the strain localization occurred mainly near dislocation nucleation sites and GBs, as illustrated in Fig. 7(e3)–(h3). With further deformation, strain spreads more uniformly across the lattice compared with 400°C, since higher temperature promoted dislocation mobility and reduced severe local stress concentration. At ϵ = 0.16 and 0.18, although high-strain regions remain concentrated along slip bands and phase boundaries, the overall strain distribution is smoother, reflecting the balance between plastic accommodation and thermal recovery.

Fig. 8(e)–(h) plotted the microstructure evolution during deformation at 600°C on the (111) crystal plane. At the early deformation stage, SFs have been readily nucleated on {111} slip planes, assisted by the low stacking fault energy of the alloy. The SFs appeared along the {111} slip planes served as precursors for partial dislocation activity. The strong slip activity on the {111} planes promoted further SFs generation, which accelerated the growth of SFs. Despite the increased SFs fraction at higher strain, some recovery processes, such as cross-slip and limited dislocation annihilation, helped accommodate strain at this advanced deformation stage.

In the current work, the MD simulations were designed as a simplified representative model to capture the fundamental deformation response of the alloy at the atomic level. Due to the limitations in computational scale and simulation timescale inherent to MD methods, it remains challenging to fully construct the experimentally observed multiphase heterogeneous microstructure, including realistic Cu-rich segregation morphology and long-range diffusion behavior. Hence, the experimentally observed Cu-rich phase evolution and heterogeneous deformation behavior were primarily analyzed through the microstructural characterization results.

In summary, the deformation mechanisms of Fe1.1NiCr1.2CoCu0.9 HEAs at elevated temperatures were dominated by the synergistic effect of dislocation activity and SFs (Fig. 9). It is characterized by active dislocation nucleation at GBs, extensive SF thickening within grains, and strong dislocation-GB interactions. The phase transition progressed rapidly through SFs thickening, while dislocations evolved from mobile partials to complex networks with high density and entanglement. GBs served as both dislocation sources and barriers, further facilitating phase evolution. In general, DRV is the rearrangement and annihilation of dislocations through processes such as climb and cross-slip, which relieves internal stress without completely forming new grains. DRX is a softening mechanism in which new strain-free grains nucleate and grow during deformation, thereby reducing dislocation density and improving ductility. The dislocation glide was easier and more

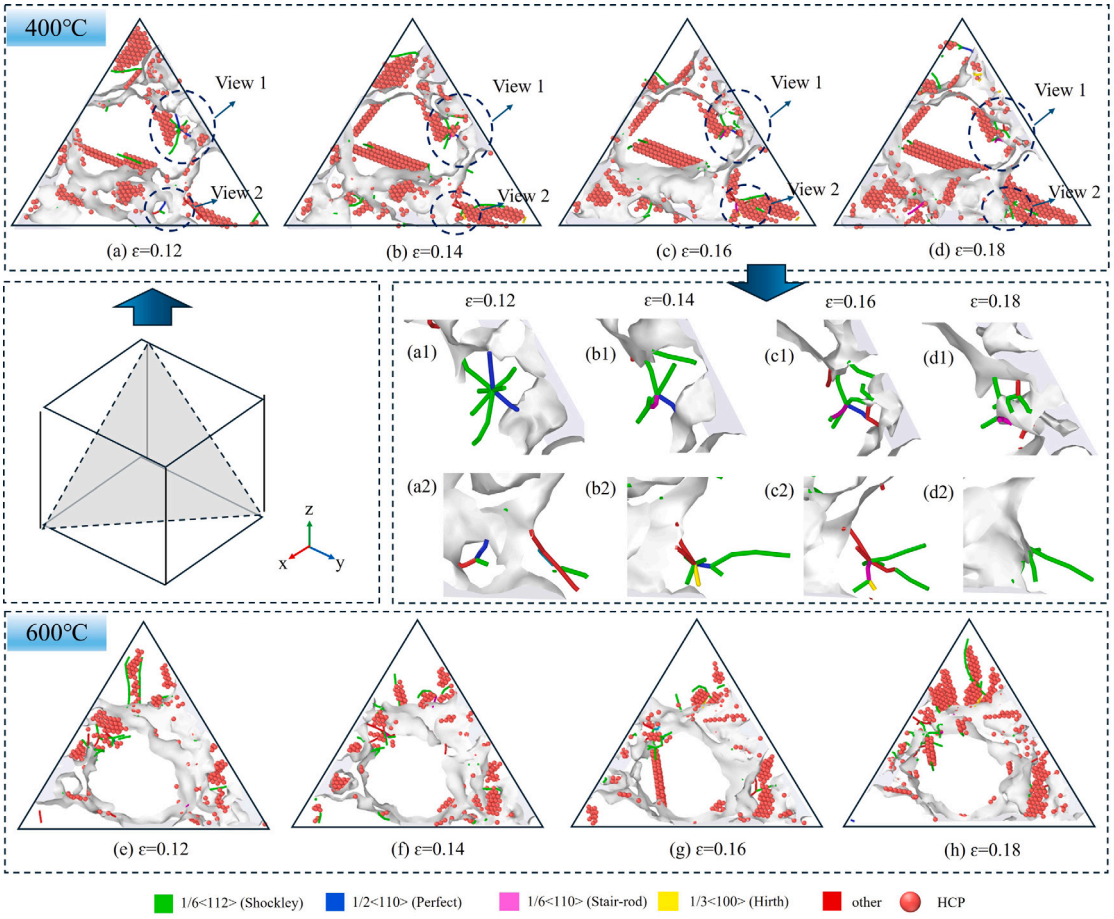


Fig. 8. (a)–(d) Microstructure evolution of Fe1.1NiCr1.2CoCu0.9 HEAs during deformation at 400°C on the (111) crystal plane: (a) ϵ = 0.12 (b) ϵ = 0.14 (c) ϵ = 0.16 (d1) ϵ = 0.18; (a1)–(d1) is the magnified perspective of the circle region View 1; (a2)–(d2) is the magnified perspective of the circle region View 2; (e)–(h) Microstructure evolution of Fe1.1NiCr1.2CoCu0.9 HEAs during deformation at 600°C on the (111) crystal plane: (e) ϵ = 0.12 (f) ϵ = 0.14 (g) ϵ = 0.16 (h) ϵ = 0.18.

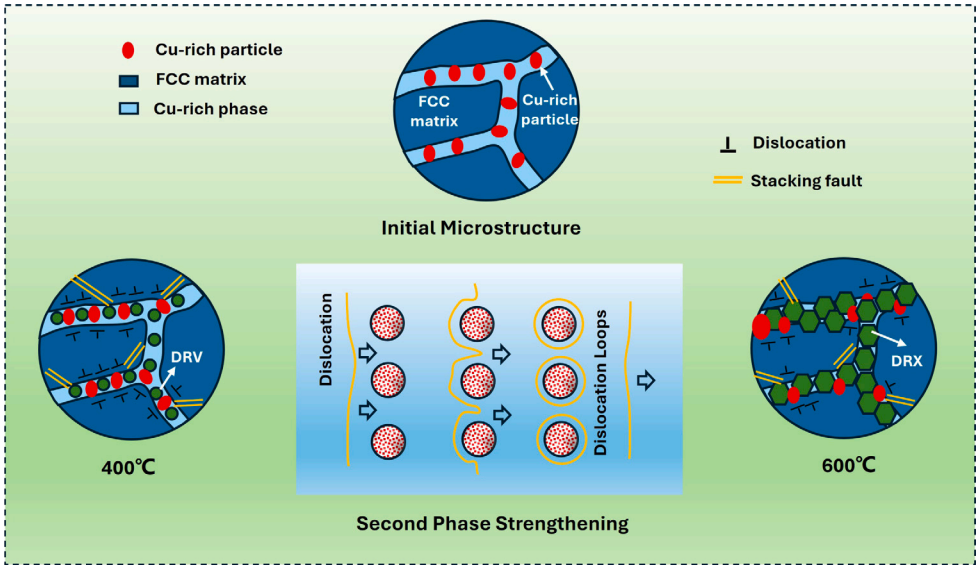


Fig. 9. Deformation mechanism of Fe1.1NiCr1.2CoCu0.9 HEAs at elevated temperatures.

extended at 600°C compared with 400°C, recovery processes were more active. Elevated temperature enhanced dislocation mobility and SF growth, but also promoted sessile dislocation formation and defect accumulation at GBs, leading to heterogeneous strain accommodation and localized microstructural instability.

4. Conclusion

In this study, a DL-based approach combining with experiments and MD simulations was proposed for composition optimization in of FeNiCrCoCu based HEAs. Fe1.1NiCr1.2CoCu0.9 HEA was designed, the microstructure and mechanical properties were investigated. The atomic scale studies were performed to explore plastic behavior by means of MD simulations. The following conclusions were derived.

(1) The developed DL model offered insights into the complex relationships between alloy composition and mechanical properties at elevated temperatures. It provided accurate predictions and (Fe, Cr) are critical in enhancing the high temperature strength, facilitating efficient alloy design.

(2) The designed Fe1.1NiCr1.2CoCu0.9 HEA is composed of dual FCC phases with the grain size of 20.3 μm. Microstructure studies showed a dendritic structure, the matrix phase was composed of nearly nominal contents of Co, Cr, Fe and Ni, whereas the interdendrites were Cu-rich phases.

(3) The fracture strength can achieve 790 MPa at 400°C, with a fairly high plastic strain of 48.6%. The strength decreased to 446 MPa as the temperature increased to 600°C. Work hardening plays a significant role in the process of plastic deformation. The Cu-rich phase, distributed at GBs, served as a preferential site for crack nucleation due to its weaker bonding with the Fe–Ni–Cr–Co matrix. It showed Cu segregation-induced interface weakening and quasi-cleavage fracture.

(4) EBSD and TEM results on deformed alloy have shed light on the underlying mechanisms that govern the high temperature mechanical properties. Dynamic recovery occurred at 400°C and DRX grains nucleated at 600°C. Deformation at 400°C is dominated by high dislocation density and strong interactions with fine Cu-rich precipitates. The recovery and precipitate coarsening at 600°C reduced defect accumulation, leading to more homogeneous deformation with improved plasticity.

(5) MD elucidates the atomic-scale mechanisms underlying macroscopic mechanical behavior, providing detailed insights into the synergistic effect of dislocation activity, extensive SF thickening within grains, and strong dislocation-GB interactions.

CRediT authorship contribution statement

Ling Qiao: Writing – original draft, Methodology, Investigation. **Gongzhuang Peng:** Investigation. **Wei Wu:** Writing – review & editing. **Jingchuan Zhu:** Validation. **George Q. Huang:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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